Brazilian E-Commerce Public Dataset

1. Dataset Understanding Report

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1. Project Background

- Source: Olist Brazilian E-Commerce Public Dataset, released on Kaggle
- Covers orders placed between September 2016 and August 2018 across all Brazilian states.
- Olist is Brazil's largest department-store marketplace aggregator, connecting small and medium businesses to major online channels.
- Dataset is structured as a relational star schema with one central orders table surrounded by customer, product, payment, review, and geolocation dimensions.

2. Raw Dataset Inventory

- Nine CSV files form the raw source layer, covering the full order lifecycle from placement to customer review.

Table Name	Raw Rows	Cols	Primary Entity
olist_orders_dataset	99,441	8	Order header and timestamps
olist_customers_dataset	99,441	5	Customer identity and location
olist_order_items_dataset	112,650	7	Line items per order
olist_order_payments_dataset	103,886	5	Payment method and value
olist_order_reviews_dataset	99,224	7	Customer ratings and comments
olist_products_dataset	32,951	9	Product catalogue and dimensions
olist_sellers_dataset	3,095	4	Seller location
olist_geolocation_dataset	1,000,163	5	Zip-code lat/lng lookup
product_category_name_translation	71	2	Portuguese to English category map

- orders and customers share identical row counts (99,441) -- a clean 1-to-1 mapping at the order level.
- order_items (112,650) outnumber orders (99,441), confirming multiple items exist per order in some transactions.
- geolocation dwarfs every other table at 1 million rows -- heavy deduplication is required before it is usable for joins.
- reviews (99,224) fall short of orders by ~217 -- a small share of completed orders carry no review record.

3. Table Relationships

- orders is the central fact table. Every other table links to it through order_id or customer_id.
- customers links to orders via customer_id. A single person may hold multiple customer_id values tracked by the stable customer_unique_id field.
- order_items links to orders via order_id, to products via product_id, and to sellers via seller_id.
- payments links to orders via order_id -- multiple payment rows can exist per order for split payments or installment records.
- reviews links to orders via order_id -- typically one review per order.
- geolocation links to customers and sellers via zip_code_prefix after deduplication.
- product_category_name_translation is a 71-row lookup joining to products on product_category_name.

4. Key Column Reference

Column	Table	Type	Business Meaning
order_id	orders	str	Unique order PK -- links all transactional tables
customer_unique_id	customers	str	Real person identity, stable across repeat purchases
order_status	orders	str	8 states: delivered, shipped, canceled, unavailable, etc.
order_purchase_timestamp	orders	datetime	Moment the customer placed the order

order_delivered_customer_date	orders	datetime	Actual delivery date; null for undelivered orders
order_estimated_delivery_date	orders	datetime	Promised delivery date for lateness measurement
price	order_items	float	Unit product price in BRL
freight_value	order_items	float	Shipping cost per line item in BRL
payment_type	payments	str	credit_card / boleto / voucher / debit_card
payment_installments	payments	int	Number of installment splits chosen by customer
review_score	reviews	int	Customer rating 1 (worst) to 5 (best)
product_category_name_english	products/tr	str	English product category after translation join
seller_state	sellers	str	2-letter state code of the seller location
geolocation_lat / lng	geolocation	float	Mean coordinates per zip code after deduplication

5. Dataset Scope and Constraints

- Temporal scope: September 2016 to August 2018. No data beyond this window.
- Geographic scope: All 27 Brazilian states represented, but demand and supply are concentrated in the southeast (SP, RJ, MG).
- Currency: All monetary values are in Brazilian Real (BRL). No conversion applied.
- All customer and seller IDs are hashed UUIDs -- the dataset is fully anonymised.
- Product names and descriptions are absent -- only category labels, dimensions, and weights are available for product-level analysis.